

# Mustakim Shikalgar

☎ +1 480-328-4635 | ✉ shikalgar.mustakim@gmail.com | 🌐 /mustakim-shikalgar | 📺 /MustakimFS

## EDUCATION

### Arizona State University

Master of Science in Software Engineering; GPA: 3.75/4.0

Tempe, AZ

Expected May 2026

## PROFESSIONAL EXPERIENCE

### Software Engineering Lead

METY Legal AI Platform (Industry Capstone) - MyEdmaster

Jan 2026 – Present

Arizona State University

- Led **6-person team** architecting production AI legal assistant serving stakeholders across **20+ legal practice domains**
- Designed **6-stage agentic RAG pipeline**: document anonymization, context-preserving summarization, intelligent clarification routing, legal reasoning engine, jurisdiction-aware formatting, impartiality verification
- Built personalized contract analysis system identifying hidden caveats, asymmetric obligations, jurisdictional risks; generates impartial educational materials with **cross-document precedent analysis**
- Implemented context-aware legal document parser extracting entity relationships, temporal dependencies, conditional clauses; integrates precedent analysis maintaining consistency across **10,000+ interactions**
- Engineered intelligent LLM routing achieving **\$0.0008/query (85% reduction, LangSmith-validated)** through dynamic model selection balancing cost, accuracy, latency across GPT-3.5, GPT-4, fine-tuned models
- Designed token-aware conversation memory with rolling summarization, self-assessment scoring, multi-turn context threading maintaining coherence across sessions
- Architected hybrid backend (**Django for application layer, FastAPI for AI/LangGraph microservices, React/TypeScript, MongoDB**) with streaming responses, concurrent request handling, comprehensive observability

## PROJECTS

### Distributed Key-Value Store | Java, Raft Consensus, gRPC, Protocol Buffers

Jan 2025 – Dec 2025

- Implemented fault-tolerant distributed database using **Raft consensus** across **5-node cluster** with leader election, log replication, linearizable consistency guarantees
- Engineered gRPC RPC layer with automatic leader discovery, client-side caching; achieved **sub-10ms p99 latency** through lease-based optimization and tunable consistency modes (CP vs AP trade-offs)

### Semiconductor Yield Prediction | Python, scikit-learn, XGBoost, Streamlit, Docker

Aug 2025 – Dec 2025

- Built production classifier achieving **76% recall, 0.81 ROC-AUC** on imbalanced SECOM dataset (**1,567 samples, 590 features, 14:1 ratio**) optimized for minimizing false negatives in manufacturing QA
- Engineered feature selection pipeline (L1 regularization, mutual information, recursive elimination) reducing dimensionality **590→113 features**; deployed Streamlit app with tunable thresholds, SHAP explanations, A/B testing

### Missing Persons Knowledge Graph | React, FastAPI, RDFLib, SPARQL

Aug 2024 – Dec 2024

- Architected semantic knowledge graph integrating **10,000+ NamUs records** using RDF ontology; enables complex relationship queries impossible with relational schemas
- Designed SPARQL optimization replacing managed GraphDB (**\$50/mo→\$0**) with FastAPI + in-memory RDFLib achieving **sub-100ms queries; published IEEE COMPSAC 2025 (27% acceptance)** demonstrating 3x improvement in case correlation

## TECHNICAL SKILLS

**Languages:** Python, Java, TypeScript/JavaScript, C++, SQL

**AI & ML:** LangChain, LangGraph, LangSmith, PyTorch, scikit-learn, RAG, Fine-tuning, Prompt Engineering

**Backend & Systems:** FastAPI, Django, Spring Boot, gRPC, Raft Consensus, Distributed Systems

**Data & Tools:** MongoDB, PostgreSQL, Redis, Neo4j, Docker, Kubernetes, AWS, Kafka, Streamlit

## ACHIEVEMENTS

**LeetCode:** 900+ problems solved (Top 15% globally), 130+ Hard including graph algorithms and system design

**Publication:** IEEE COMPSAC 2025 (27% acceptance) - knowledge graph for missing persons tracking